ENGINEERING TOMORROW

User Manual

Remote Control
RCT Rechargeable Batteries

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Revision history

Table of revisions

Date	Changed	Rev
February 2025	Revised information on batteries and safety	0101a
November 2023	Initial Version	0101x

Contents

Battery Models and Their Technology

Charger and battery

Disposal note:

This symbol on the product indicates that it may not be disposed of as household waste.

It must be handed over to the applicable take-back scheme for the recycling of electrical equipment.

- Dispose of the product through channels provided for this purpose.
- Comply with all local and currently applicable laws and regulations.

Battery models and their Technology

Danfoss RCT Remote control Transmitters do combine different Battery models according to capacity and Technology.

This Manual is meant to show the different Battery technologies, and the recommendations for storing batteries as well as when using them.

RCT transmitters do use NiMH, Li Ion and alkaline Batteries depending on the models.

NiMH Batteries

A **nickel–metal hydride battery** (**NiMH** or **Ni–MH**) is a type of rechargeable battery. The chemical reaction at the positive electrode is similar to that of the nickel-cadmium cell (NiCd), with both using nickel oxide hydroxide (NiOOH). However, the negative electrodes use a hydrogen-absorbing alloy instead of cadmium. NiMH batteries can have two to three times the capacity of NiCd batteries of the same size, with significantly higher energy density, although only about half that of lithium-ion batteries.

They are typically used as a substitute for similarly shaped non-rechargeable alkaline batteries, as they feature a slightly lower but generally compatible cell voltage and are less prone to leaking.

The battery models used on RCT Transmitters are BT06K and BT27IK and for ATEX Transmitters the BT06K ATEX battery is used.

Li Ion Batteries

A **lithium-ion** or **Li-ion battery** is a type of rechargeable battery which uses the reversible intercalation of Li+ ions into electronically conducting solids to store energy. In comparison with other rechargeable batteries, Li-ion batteries are characterized by a higher specific energy, higher energy density, higher energy efficiency, longer cycle life and longer calendar life.

The battery model used for RCT transmitters is the BT11K Li Ion Battery.

Alkaline Batteries

An **alkaline battery** (IEC code: L) is a type of primary battery where the electrolyte (most commonly potassium hydroxide) has a pH value above 7. Typically these batteries derive energy from the reaction between zinc metal and manganese dioxide.

Alkaline Batteries supplied with the remote control are Disposable and Not Rechargeable. Depending on the Transmitter the Battery cell size will be AA or AAA.

On the next chapters we will see the Rechargeable Batteries main characteristics, as well as the relation between transmitters and batteries, followed by the Battery charging and storing recommendations. Finally we will check the Battery chargers used to charge the different models.

Rechargeable Batteries Main Characteristics

Rechargeable Batteries main characteristics

Battery Model	BT06K	BT27IK	BT11K	BT06K ATEX
Technology	NiMH	NiMH	Li Ion	NiMH
Capacity (mAh)	600mAh	2700mAh	1130mAh	600mAh
Voltage (Vdc)	4,8 Vdc	4,8 Vdc	3,7 Vdc	4,8 Vdc
Weight (gram)	70,3 g	120 g	23 g	75 g
Discharge Temperature	-20ºC to 50ºC	-20ºC to 55ºC	-20ºC to 50ºC	-15ºC to 70ºC
Charging Temperature Range	10ºC to 45ºC	0ºC to 40ºC	0ºC to 45ºC	0ºC to 50ºC
Charging Cycles	500	500	500	500
Storage Temperature	15ºC to 25ºC	15ºC to 25ºC	15ºC to 25ºC	15ºC to 25ºC
Storage Temperature (1 year)	-20ºC to 35ºC	-20ºC to 35ºC	-20ºC to 35ºC	-20ºC to 30ºC
Storage Relative Humidity	+65% +-20%	+65% +-20%	+65% +-20%	+65% +-20%
Recharge Time Lapse	3 months	3 months	3 months	3 months
Minimum Charge every time	100%	100%	40%	100%

Transmitters vs Rechargeable Battery model

Battery model	Battery Technology	Transmitter Models
BT06K	NiMH Rechargeable	T70/1, T70/2, T70/1 HALL, T70/2 HALL
BT27IK	NiMH Rechargeable	T IK3, T IK4, T Ikovet
BT11K	LiIon Rechargeable	T IkoreB, T Ikargo1, T Ikargo2, T IK1-G, T IK2
BT06K ATEX	NiMH Rechargeable	T70/1 ATEX, T70/1 ATEX CSA, T70/2 ATEX, T70/2 ATEX CSA
AAA	Alkaline	T Ikore, T Ikompact8, T Ikompact12
AA	Alkaline	T IK1-G

Battery Certifications

Battery Certifications:

- BT11K:
  - CE Marking (EMC directive 2014/30/EU),
  - IEC 62133-2:2017,
  - IEC 62281: 2019,
  - PSE Marking (METI standards),
  - RCM Marking (ACMA standards)
  - PCT Marking (IEC 61960:2007)
  - RoHS Directive 2011/65/EU and (EU) 2015/863
  - UL2054 (List, product code E523397)

Rechargeable Batteries Main Characteristics

  - UL 60950-1 (List, product code E523397)
  - CAN/CSA-C22.2 No. 60950-1 (List, product code E523397)
  - UN 38.3
- BT27K:
  - CE Marking
  - IEC 62133-1: 2017
- BT06K-ATEX:
  - CE Marking

Maintenance and good practice Recommendations

Battery Charging Recommendations

Charge the battery fully before use. This ensures that the battery's full capacity will be available. The battery lifespan is estimated to 500 recharging cycles and is largely dependent on the conditions of use. To maximize the lifespan of the batteries and battery charger, follow these recommendations:

- Do not recharge the battery until it is completely flat, as shown with red LED slow pulse on the transmitter
- Always charge the batteries at temperatures between 0° and 45°C (the batteries will not become fully charged at temperatures exceeding 45°C)
- Do not leave the battery charger or batteries in a direct sunlight
- Charge batteries at least once every three months
- Make the charge of at least 40% of the full charge.
- Ideal Battery storage temperature should be between 15ºC and 25ºC.
- Avoid short circuits between the battery contacts; do not carry charged batteries in toolboxes or next to other metal objects (keys, coins, etc.)
- Always keep contacts clean
- Caution! Risk of Explosion if Battery is Replaced by an incorrect type. Non Danfoss Battery use may void warranty

Battery Storage Recommendations

The following recommendations should be kept in mind when storing NiMH or Li-ion rechargeable batteries:

- Always charge the battery completely before initial use.
- For storing Li-ion batteries, a battery charge of 30 – 50 % is ideal. Charge an empty battery for approx. 1 hour to reach this level of charge.
- NiMH batteries should ideally be 100 % charged when you store them. This helps to prevent total discharging of the battery.
- Rechargeable batteries lose capacity that cannot be recovered when stored for long periods.
- Therefore, charge the batteries at intervals not exceeding 3 months.

Tips when using the Battery

Here we do provide some helpful instructions to allow you to get the maximum benefit from our batteries for as long as possible:

- Please note the specified temperature ranges for using, charging, and storing your Rechargeable batteries (see Batteries main characteristics). The use, charge and storage of the batteries at extreme temperatures outside of the specified range can compromise their capacity.
- Always charge your batteries in a timely manner and avoid your batteries to fully discharge, as this will cause irreparable damage.
- Never store your batteries when they are discharged; always ensure that NiMH batteries are fully charged (100 %) before storing them, and LiIon Batteries with a level of 30 – 50 % is ideal.

Battery Transport

Adhere to the legal regulations for the transport and shipping of NiMH or Li-ion batteries applicable to your country. We deliver all Danfoss batteries with Li-ion technology in a suitable packaging with all required documents and safety notifications.

In the case of air transport, the charge level of the batteries must not exceed 30 percent.

For this reason, we charge our Li-ion batteries to a level of 20 to 30 percent before dispatch. This also allows for an extended storage period that will not result in a reduction of the service life.

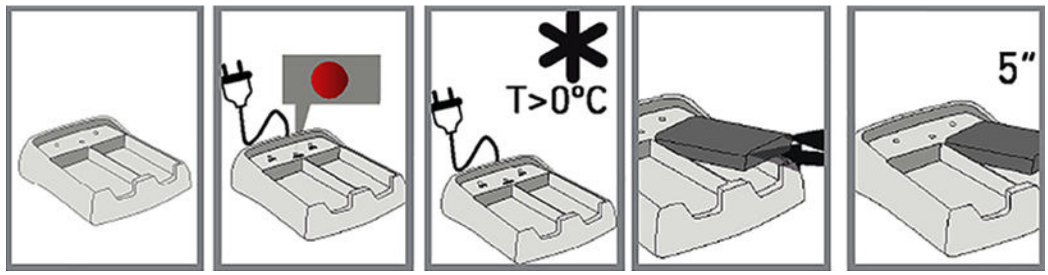
Maintenance and good practice Recommendations

Before transporting or sending the batteries, make sure to properly protect the battery contacts from short circuiting.

Associated Battery Chargers

BC70K Battery Charger Set-up

The battery charger has two charging compartments that can simultaneously charge two BT11K batteries. Use the information below to set up the BC70K battery charger.



1. Connect the charger to a power source using the provided power supply.
   The red LED will switch on if the charger is properly connected.
2. Place the batteries on the charger.
3. Optional: When charging two batteries, wait at least five seconds before inserting the second battery into the compartment.

Warning

Possible damage to battery.
The Battery Charger must be installed in a dry/interior environment. Make sure to charge batteries in environments with temperatures over 0°C.

BC70K Charger LEDs Status

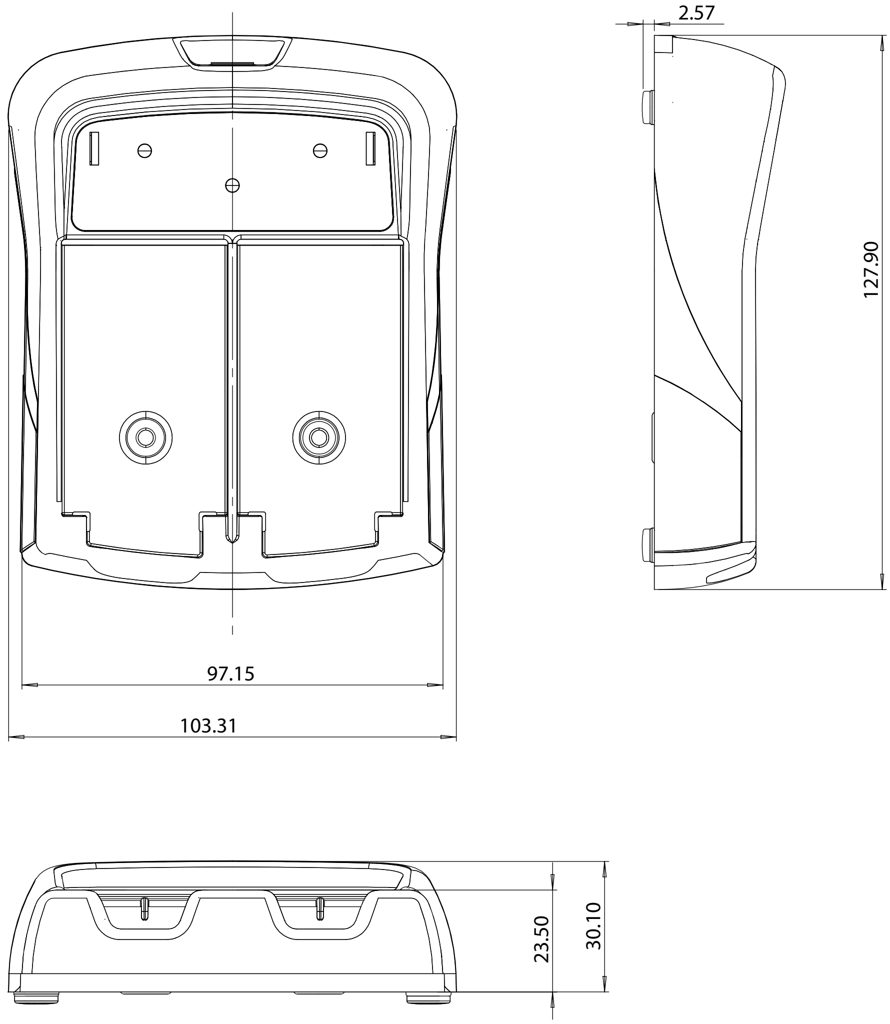
The BC70K charger has a LED for each compartment (**BAT 1** and **BAT 2**) and a common indicator (**POWER**).

LED color / frequency	Description
Green LED / pulsing (BAT 1, BAT 2)	The battery is being charged
Green LED / continuous (BAT 1, BAT 2)	The battery is completely charged
Red LED / pulsing or continuous (BAT 1, BAT 2)	The battery charger fault
Red LED / continuous (POWER)	The charger is properly connected to power source

Associated Battery Chargers

BC70K Battery Charger Dimensions

Dimensions (mm)

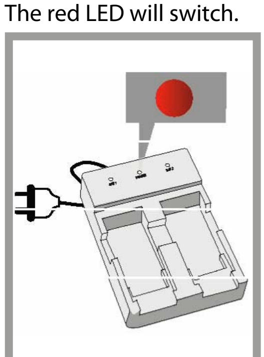


Setting up the CB70 battery charger

Use the information below to properly set up the CB70 batter charger.

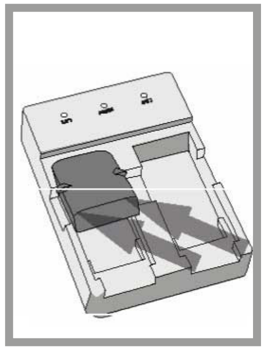
The battery charger has two charging compartments that can simultaneously charge two batteries.

1. Connect the charger to a power source using the cable supplied.
   The red LED will switch.



Associated Battery Chargers

2. Place the batteries in the compartments of the battery charger.



3. Optional: If charging multiple batteries, wait at least 5 seconds before placing the second battery in the other compartment.

Possible damage to battery!
The Battery Charger must be installed in a dry/interior environment. Make sure to charge batteries in environments with temperatures over 0° C.

CB70 status LEDs

Each battery compartment has an LED that indicates the status of the batteries' charge.

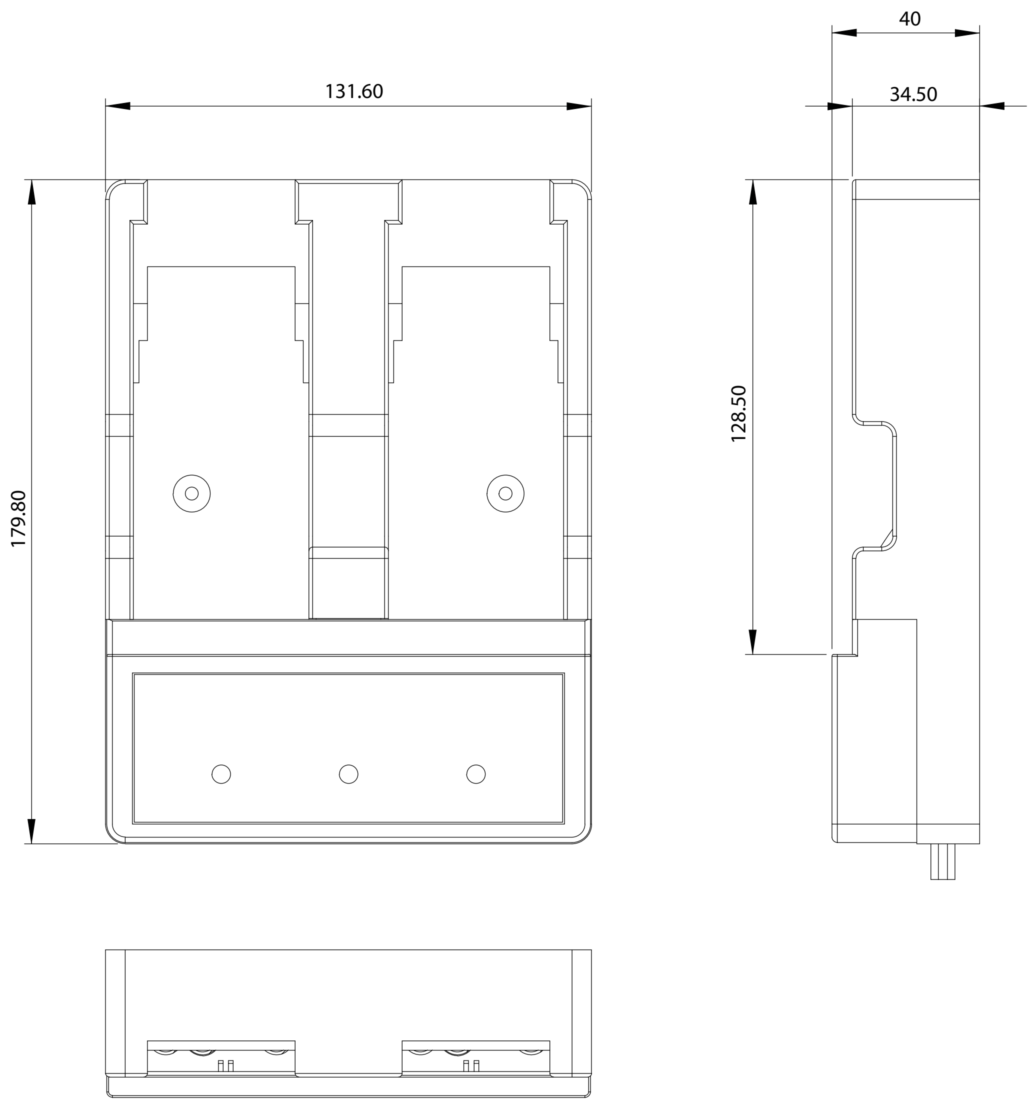
Green LED; pulsing	Battery is excessively depleted
Green LED; continuous	Normal charging operation mode
Green LED; off	Battery charging process is complete

The battery charger must be placed and used out of the danger area.

Associated Battery Chargers

CB70 battery charger dimensions

Dimensions in mm



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